ICT2142 · Database Systems

Practical Tutorial: Transactions and Query Safety

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

A transaction groups a consistent set of database changes. COMMIT makes the changes durable; ROLLBACK abandons them. Isolation controls how concurrent transactions observe intermediate states. Use prepared statements for values and least-privilege database accounts.

2. Example and reasoning

When publishing an assessment plan, insert all components inside one transaction and verify the total weight before committing. If validation fails, roll back the entire plan rather than leaving a partial set.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “practical tutorial: transactions and query safety” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.